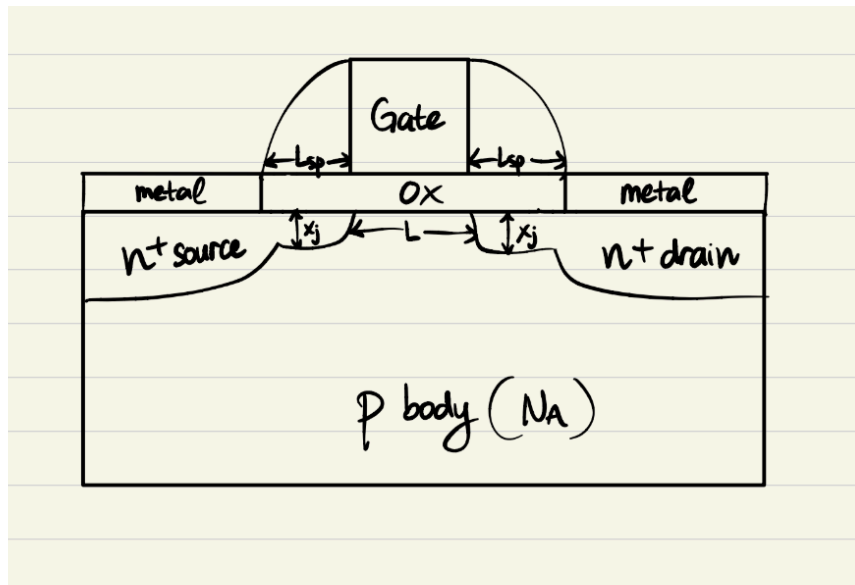


ELENG-130/230A Final Project

Due 12/12/2025

1) MOSFET Design



Final design parameter values:

$$L_{SP} = 0.020 \text{ nm}$$

$$N_A = 1 \times 10^{18} \text{ cm}^{-3}$$

$$X_j = 0.007 \text{ nm}$$

Final device current properties:

$$I_{ON} = 405.4 \text{ } \mu\text{A}$$

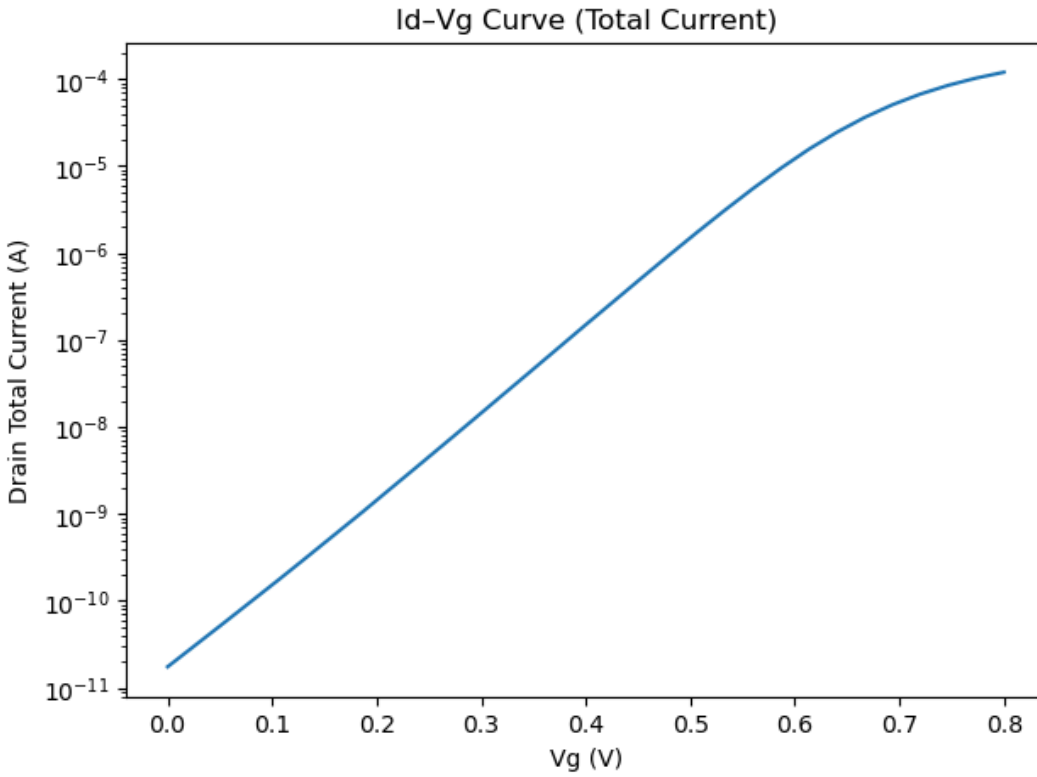
$$I_{OFF} = 0.9544 \text{ nA}$$

We performed a sweep for N_A and ultimately chose N_A to be $1 \times 10^{18} \text{ cm}^{-3}$. We then swept X_j in 4nm increments and decided on $X_j = 8\text{nm}$. We then swept L_{SP} across the allowed range and found that our device was very close to meeting the design requirements. We then did a finer sweep around $X_j = 8\text{nm}$ (only varying it by 1nm) and varied L_{SP} the same way. We discovered our final design parameters above satisfied the device requirements.

2) I_{DS} - V_{GS} simulations

From our simulations, we found $V_{Tsat} = 0.411$ V and $V_{Tlin} = 0.545$ V.

$$DIBL = \frac{|V_{Tlin}| - |V_{Tsat}|}{V_{DD} - V_{dlin}} = \frac{|0.545| - |0.411|}{0.8 - 0.05} \approx 0.179 \text{ V/V} = 179 \text{ mV/V}$$



$$I_1 = 1.020739 \times 10^{-10}$$

$$V_{G1} = 0.09044248 \text{ V}$$

$$I_2 = 1.038362 \times 10^{-9}$$

$$V_{G2} = 0.1868142 \text{ V}$$

$$\frac{1}{SS} = \frac{\partial \log_{10}}{\partial V_{GS}} = \frac{\log_{10}(1.038362 \times 10^{-9}) - \log_{10}(1.020739 \times 10^{-10})}{0.1868142 - 0.09044248} = 10.45363$$

$$SS = \frac{1}{10.45363} = 0.0956606 \text{ V/dec} = 95.6606 \text{ mV/dec}$$

3) Threshold voltage (V_T)

$$V_T = \phi_{MS} + 2\phi_F + \frac{\sqrt{2\epsilon_{si}qN_A(2\phi_F)}}{C_{ox}}$$

$$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{4\epsilon_{si}qN_A\phi_F}}{C_{ox}}$$

$$N_A = 1.0 \times 10^{18} \text{ cm}^{-3}$$

$$T_{ox} = 1 \text{ nm} = 1 \times 10^{-7} \text{ cm}$$

$$\Phi_M = 4.6 \text{ eV}$$

$$\phi_F = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) = 0.0259 \ln\left(\frac{1 \times 10^{18}}{1 \times 10^{10}}\right) = 0.477096 \text{ V}$$

$$\Phi_S = \chi + \frac{E_g}{2} + \phi_F = 4.05 + \frac{1.12}{2} + 0.477096 = 5.087096 \text{ eV}$$

$$V_{FB} = \Phi_M - \Phi_S = 4.6 - 5.087096 = -0.487096 \text{ V}$$

$$Q_{dep} = \sqrt{4\epsilon_{si}qN_A\phi_F}$$

$$Q_{dep} = \sqrt{4(11.8 \times 8.854 \times 10^{-14})(1.6 \times 10^{-19})(10^{18})(0.477096)}$$

$$Q_{dep} = 5.64812 \times 10^{-7}$$

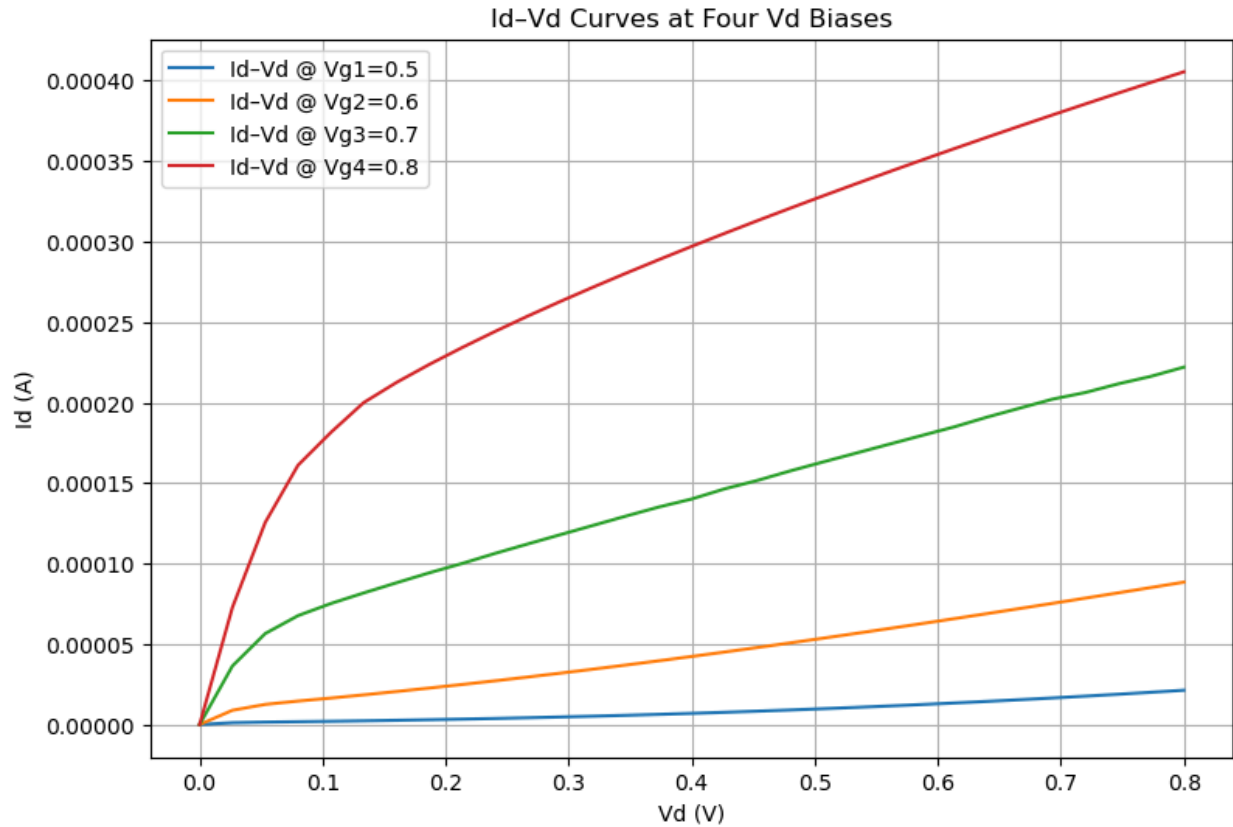
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.854 \times 10^{-14}}{1 \times 10^{-7}} = 3.45306 \times 10^{-6} \text{ F/cm}^2$$

$$V_T = V_{FB} + 2\phi_F + \frac{Q_{dep}}{C_{ox}}$$

$$V_T = -0.487096 + 2(0.477096) + \frac{5.64812 \times 10^{-7}}{3.45306 \times 10^{-6}} = 0.630665 \text{ V}$$

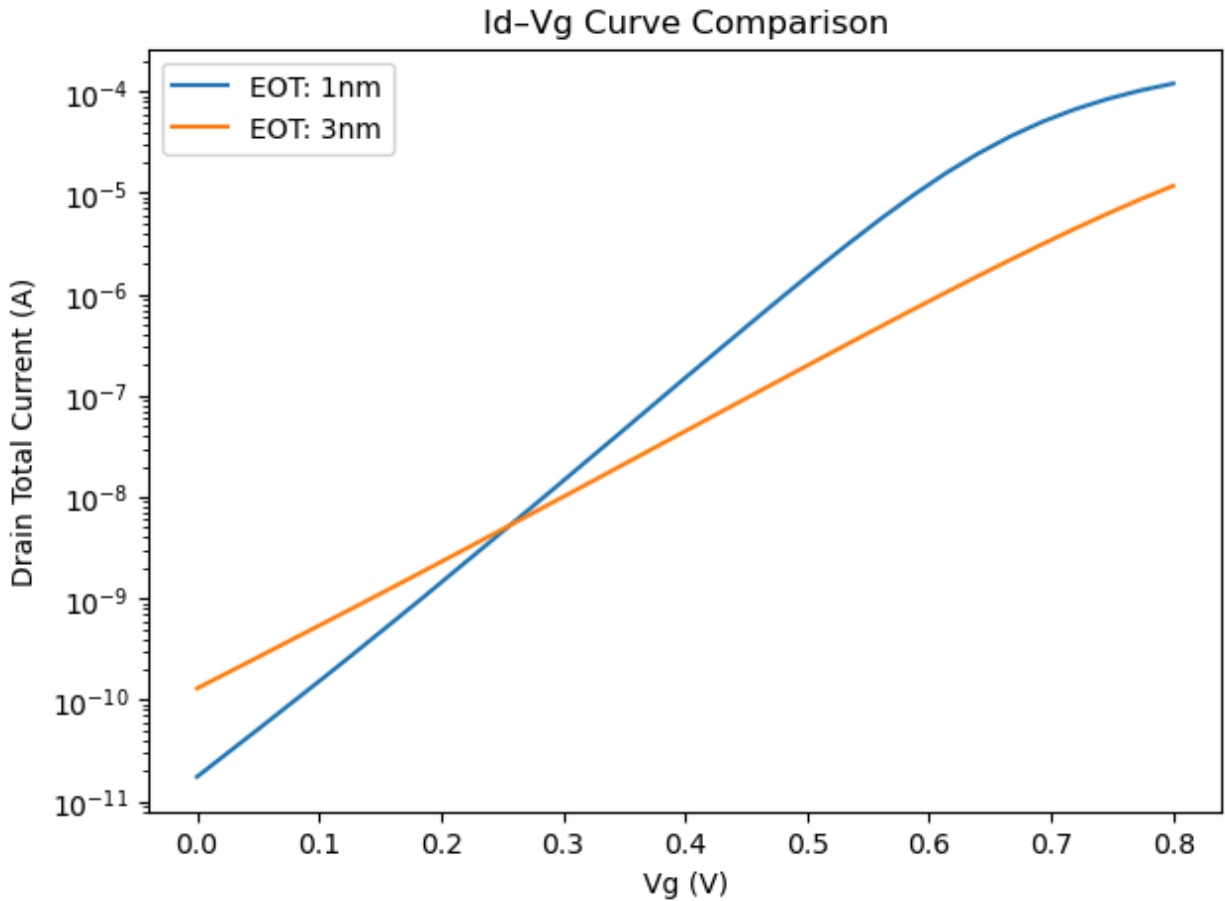
In the simulation, the extractor gives $V_T \approx 0.545 \text{ V}$ in the linear region and a smaller $V_T \approx 0.411 \text{ V}$ in saturation. The MOSCAP calculation gives a larger $V_T \approx 0.63 \text{ V}$ because it assumes an ideal long-channel MOS capacitor with $V_{DS} = 0$ and ignores short-channel effects and DIBL, which lower V_T in the simulated MOSFET.

4) I_{DS} - V_{DS} simulations



In the I_d - V_d plot, it doesn't show that the MOSFET is exhibiting square-law behavior. The I_d - V_d curves of this device increase almost linearly with V_{GS} , and the saturation region flattens early. This behavior is expected in short-channel MOSFETs, where velocity saturation, mobility degradation, and strong electric fields dominate transport. Because the device is operating in the velocity saturated regime rather than the long-channel quadratic regime, the square-law dependence does not apply.

5) Effective Oxide Thickness (EOT)



When T_{ox} is increased from 1 nm to 3 nm, the Id-Vg curve shifts to higher V_{GS} , the saturation current decreases, and off-state leakage increases due to poorer gate control and stronger short-channel effects.

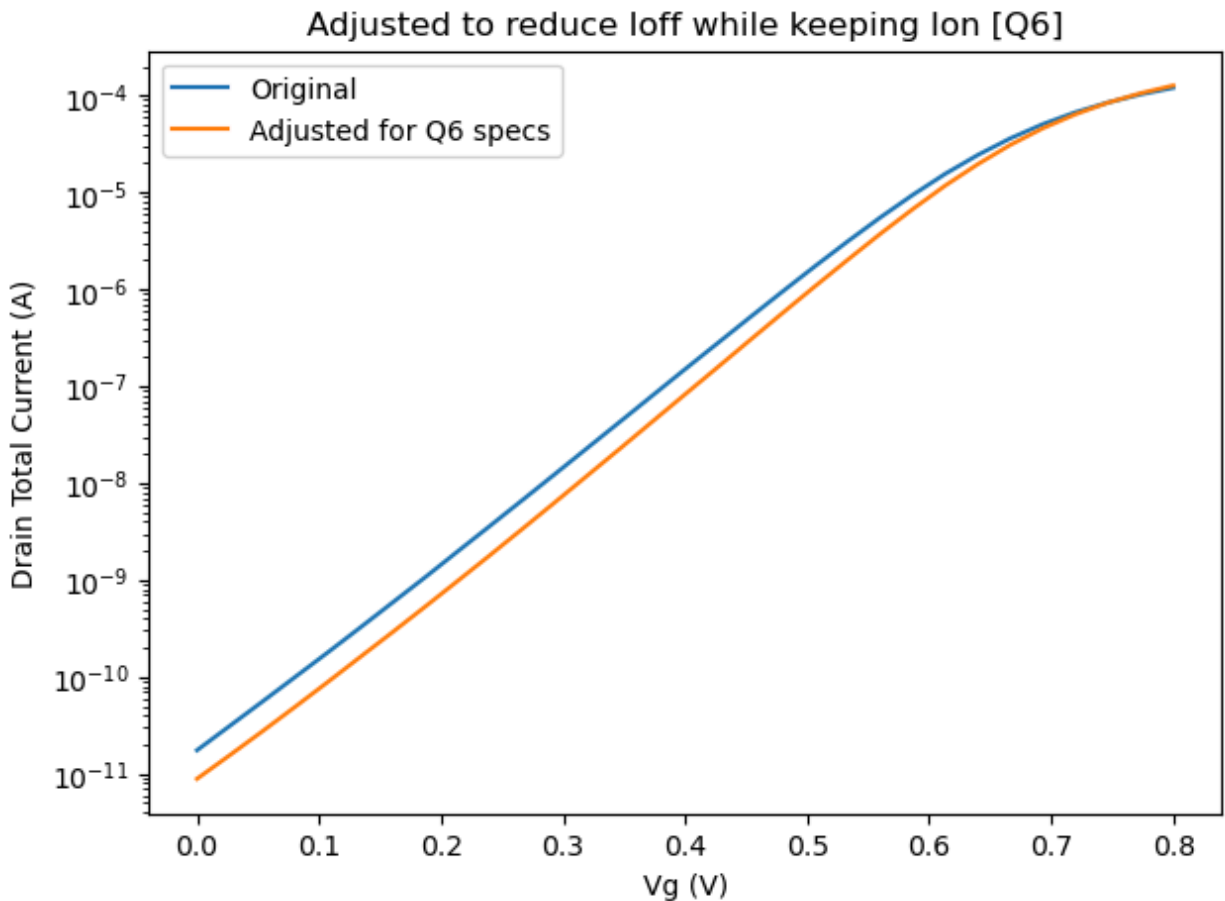
For future technology generations with even shorter gate lengths, the effective oxide thickness must remain very thin to maintain good electrostatics and suppress DIBL and leakage. This is why advanced technologies rely on low-EOT metal-gate instead of using thicker SiO_2 .

6) Increasing ON/OFF ratio

To reduce the OFF current significantly while keeping the ON current nearly intact, we can adjust the gate work function and the oxide thickness. Slightly increasing the gate work function will raise the threshold voltage of the MOSFET, which exponentially reduces the subthreshold leakage current by making it harder for carriers to flow when the device is in the OFF state. At the same time, decreasing the effective oxide thickness increases the gate capacitance, which strengthens the gate's electrostatic control over the channel. This improves the subthreshold slope and mitigates short-channel effects like DIBL, further reducing I_{OFF} without compromising on I_{ON} . After increasing the gate work function and decreasing the oxide thickness, we can then adjust other design parameters to maintain I_{ON} at over 400 μA .

Simulation results with 4.65 eV gate work function, $T_{\text{OX}} = 0.8 \text{ nm}$, $N_{\text{A}} = 9.6 \times 10^{17} \text{ cm}^{-3}$, $L_{\text{SP}} = 17.5 \text{ nm}$, and $X_{\text{j}} = 7 \text{ nm}$ gives us $I_{\text{OFF}} = 0.6439 \text{ nA}$ (significantly smaller than before) while maintaining I_{ON} to be 405.3 μA (barely any change from the original design parameters).

See below for the updated I_{d} - V_{g} curve with the parameters listed above:



7) Increasing Temperature

As temperature increases, the MOSFET's ON current generally decreases. This happens because carrier mobility is strongly degraded by increased phonon scattering at higher temperatures.

In contrast, the OFF current increases significantly at higher temperatures. Subthreshold leakage is thermally activated, meaning that as temperature rises, more carriers have enough energy to overcome the source-channel barrier even when the gate voltage is low.

Finally, the subthreshold swing becomes worse as temperature increases. Because the subthreshold swing is proportional to kT/q , higher temperature directly increases the minimum achievable slope. As a result, the transition from OFF to ON becomes less sharp, and the device requires a larger change in gate voltage to achieve the same change in drain current.